

Experiment 2 – Exploratory Data Analysis (EDA)

Aim

To import a CSV file using the Pandas library in Python and display its contents for basic data inspection.

Procedure

1. Open Jupyter Notebook and import the Pandas library.
2. Create or place the `student.csv` file in the current working directory.
3. Use the `pd.read_csv()` function to load the CSV file into a Pandas DataFrame.
4. Store the imported data in a variable (e.g., `df`).
5. Display the DataFrame using the `print()` function.
6. Verify that the data has been imported correctly by checking the displayed records.
7. Analyze the imported dataset for further processing if required.

Program

DATA IMPORT

Algorithm - 1

```
import pandas as pd

df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")

print(df.head())

print(df)

print(df.tail())

print(df.info())

print(df.shape)

print(df.columns)

print(df.describe())
```

Output

```
In [14]: import pandas as pd

df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")

print(df.head())
print(df)
print(df.tail())
print(df.info())
print(df.shape)
print(df.columns)
print(df.describe())
```

	Location	Country	Category	Visitors	Rating	Revenue	\
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38	
1	aHKUXhjzTo	USA	Historical	813627	2.01	802625.60	
2	dlrdYtJFTA	Brazil	Nature	508673	1.42	338777.11	
3	DxmlzdGkHK	Brazil	Historical	623329	1.09	295183.60	
4	WJCCQlepzn	France	Cultural	124867	1.43	547893.24	

Accommodation_Available

0	Yes
1	No
2	Yes
3	Yes
4	No

	Location	Country	Category	Visitors	Rating	Revenue	\
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38	
1	aHKUXhjzTo	USA	Historical	813627	2.01	802625.60	
2	dlrdYtJFTA	Brazil	Nature	508673	1.42	338777.11	
3	DxmlzdGkHK	Brazil	Historical	623329	1.09	295183.60	
4	WJCCQlepzn	France	Cultural	124867	1.43	547893.24	

Algorithm – 2

```
import pandas as pd

df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")

df.to_excel(r"C:\Users\HDC0422062\Downloads\tourism_dataset.xlsx", index=False)

print("Excel file created successfully.")

print(df)

print(df.tail())

print(df.info())

print(df.shape)

print(df.columns)

print(df.describe())
```

Output

```
jupyter DATA-VISUALISATION(EXPERIMENT) Last Checkpoint: 20 minutes ago (autosaved)
File Edit View Insert Cell Kernel Widgets Help
In [18]: import pandas as pd

df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")
df.to_excel(r"C:\Users\HDC0422062\Downloads\tourism_dataset.xlsx", index=False)

print("Excel file created successfully.")
print(df)
print(df.tail())
print(df.info())
print(df.shape)
print(df.columns)
print(df.describe())

Excel file created successfully.
   Location Country Category  Visitors  Rating  Revenue \
0  kuBZRkVsAR  India  Nature    948853    1.32   84388.38
1  aHKUXhjzTo   USA  Historical  813627    2.01  802625.60
2  dlrdYtJFTA  Brazil  Nature    508673    1.42  338777.11
3  Dxm1zdGkHK  Brazil  Historical  623329    1.09  295183.60
4  WJCCQlepzn  France  Cultural   124867    1.43  547893.24
...  ...  ...  ...  ...  ...  ...
5984 xAzwnVKAqz   USA  Urban     828137    1.97  132848.78
5985 IfKotyaJFC  France  Nature   276317    3.53  325183.96
5986 bPyubCWGgA  Egypt  Beach     809198    3.37  927336.50
5987 kkWIucpBnu  Egypt  Cultural   808303    2.52  115791.43
5988 gHXUrdticm  France  Cultural    40939    4.65  957026.85

Accommodation_Available
0      Yes
1      No
2      Yes
3      Yes
4      No
...  ...
5984  No
5985  Yes
5986  No
5987  Yes
5988  Yes

[5989 rows x 7 columns]
   Location Country Category  Visitors  Rating  Revenue \
5984 xAzwnVKAqz   USA  Urban     828137    1.97  132848.78
5985 IfKotyaJFC  France  Nature   276317    3.53  325183.96
5986 bPyubCWGgA  Egypt  Beach     809198    3.37  927336.50
5987 kkWIucpBnu  Egypt  Cultural   808303    2.52  115791.43
5988 gHXUrdticm  France  Cultural    40939    4.65  957026.85
```

Algorithm – 3

```
import pandas as pd
```

```
import sqlite3
```

```
df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")
```

```
conn = sqlite3.connect("tourism.db")
```

```
df.to_sql("tourism", conn, if_exists="replace", index=False)
```

```
query = "SELECT * FROM tourism"
```

```
sql_df = pd.read_sql(query, conn)
```

```
print(sql_df)
```

```
print(sql_df.head())
```

```
conn.close()
```

Output:

```
query = "SELECT * FROM tourism"

sql_df = pd.read_sql(query, conn)

print(sql_df)

print(sql_df.head())

conn.close()
```

	Location	Country	Category	Visitors	Rating	Revenue	\
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4	WJCCQlepzn	France	Cultural	124867	1.43	547893.24	
...	...	...	...	...	...	...	
5984	xAzwnVKAqz	USA	Urban	828137	1.97	132848.78	
5985	IfKotyaJFC	France	Nature	276317	3.53	325183.96	
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	Accommodation_Available
0	Yes
1	No
2	Yes
3	Yes
4	No
...	...
5984	No
5985	Yes
5986	No
5987	Yes
5988	Yes

[5989 rows x 7 columns]

	Location	Country	Category	Visitors	Rating	Revenue	\
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38	
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4	WJCCQlepzn	France	Cultural	124867	1.43	547893.24	

Algorithm – 4

```
import pandas as pd
```

```
url = "https://raw.githubusercontent.com/srihar12/datavisualizationtechniques/main/Tourism-Analysis/tourism_dataset.csv"
```

```
df = pd.read_csv(url)
```

```
print(df.head())
```

Output:

```
: import pandas as pd

url = "https://raw.githubusercontent.com/srihar12/datavisualizationtechn

df = pd.read_csv(url)

print(df.head())
```

	Location	Country	Category	Visitors	Rating	Revenue	\
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38	
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3	DxmlzdGkHK	Brazil	Historical	623329	1.09	295183.60	
4	WJCCQlepzn	France	Cultural	124867	1.43	547893.24	

	Accommodation_Available
0	Yes
1	No
2	Yes
3	Yes
4	No

DATA EXPORT

Program

Algorithm – 1(Export to csv file)

```
In [24]: import pandas as pd

df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")

df.to_csv(r"C:\Users\HDC0422062\Downloads\output.csv", index=False)

print("CSV File Exported")

CSV File Exported
```

Algorithm – 2 (Export to excel file)

```
In [25]: import pandas as pd

df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")
|
df.to_excel(r"C:\Users\HDC0422062\Downloads\output.xlsx", index=False)

print("Excel File Exported")

Excel File Exported
```

Algorithm- 3(Export to json file)

```
] : import pandas as pd

# Read the dataset
df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")

# Export to JSON
df.to_json(r"C:\Users\HDC0422062\Downloads\output.json")

print("JSON File Exported")
```

JSON File Exported

Visualization of Imported Data

Program

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv(r"C:\Users\HDC0422062\Downloads\tourism_dataset.csv")
```

```
country_count = df["Country"].value_counts()
```

```
country_count.plot(kind="bar")
```

```
plt.title("Number of Tourist Locations by Country")
```

```
plt.xlabel("Country")
```

```
plt.ylabel("Count")
```

```
plt.show()
```

output:

